



JavaScript — Increment / Decrement Operators Assignment

Instructions

For each question, carefully analyze the execution order of the `++` and `--` operators and write the expected output or explain the behavior.

1) What will be the output of the following JavaScript program?

```
let i = 11;  
  
i = i++ + ++i;  
  
console.log(i);
```

2) Guess the output of the following program.

```
let a = 11;  
let b = 22;  
let c;  
  
c = a + b + a++ + b++ + ++a + ++b;  
  
console.log("a=" + a);  
console.log("b=" + b);  
console.log("c=" + c);
```

3) What will be the output of the below program?

```
let i = 0;  
  
i = i++ - --i + ++i - i--;  
  
console.log(i);
```

4) Is the below JavaScript program written correctly?

If it is valid, what will be the output?

```
let b = true;
```

```
console.log(b);
```

5) What will be the output of the below program?

```
let i = 1;
```

```
let j = 2;
```

```
let k = 3;
```

```
let m = i-- - j-- - k--;
```

```
console.log("i=" + i);
```

```
console.log("j=" + j);
```

```
console.log("k=" + k);
```

```
console.log("m=" + m);
```

6) What will be the output of the following program?

```
let a = 1;
```

```
let b = 2;
```

```
console.log(--b - ++a + ++b - --a);
```

7) What will be the value of **i**, **j** and **k** in the below program?

```
let i = 19;
```

```
let j = 29;
```

```
let k;
```

```
let m = i-- - j-- - k--;
```

```
console.log("i=" + i);
```

```
console.log("j=" + j);
```

```
console.log("k=" + k);
```

```
console.log("m=" + m);
```

8) What is wrong with the below JavaScript statement?

Will it execute successfully? If not, identify the reason.

```
let i = 11;
```

```
let j = --(i++);
```

9) Guess the value of `p` in the below program.

```
let m = 0;
```

```
let n = 0;
```

```
let p = --m * --n * n-- * m--;
```

```
console.log(p);
```

10) What will be the output of the following program?

```
let a = 1;
```

```
a = a++ + ++a * --a - a--;
```

```
console.log(a);
```

11) What will be the outcome of the below program?

Will it execute successfully or produce an error?

```
let a = 11++;
```

```
console.log(a);
```

12) What will be the outcome of the following JavaScript program?

```
let ch = "A";
```

```
console.log(ch++);
```

Also determine the final value of `ch`.

13) What will be the outcome of the following JavaScript program?

```
let ch = "A";
```

```
console.log(++ch);
```

Explain why the result is different from the previous question.

14) What will be the output of the following program?

```
let d = 1.5;
```

```
let D = 2.5;
```

```
console.log(d++ + ++D);
```

Also determine the final values of **d** and **D**.

Bonus Questions

15) What will be the output?

```
let x = 5;
```

```
console.log(x++);
```

```
console.log(++x);
```

```
console.log(x--);
```

```
console.log(--x);
```

16) What will be the final value of **x**?

```
let x = 10;
```

```
x = x++ + x++ + ++x;
```

```
console.log(x);
```

17) What will be the output?

```
let a = 5;
```

```
let b = a++;
```

```
console.log(a);  
console.log(b);
```

18) What will be the output?

```
let a = 5;  
let b = ++a;
```

```
console.log(a);  
console.log(b);
```

19) What will be the output?

```
let x = 10;
```

```
console.log(x++ + ++x);
```

20) What will be the final value of **x**?

```
let x = 5;
```

```
x++;  
++x;  
x--;  
--x;
```

```
console.log(x);
```
